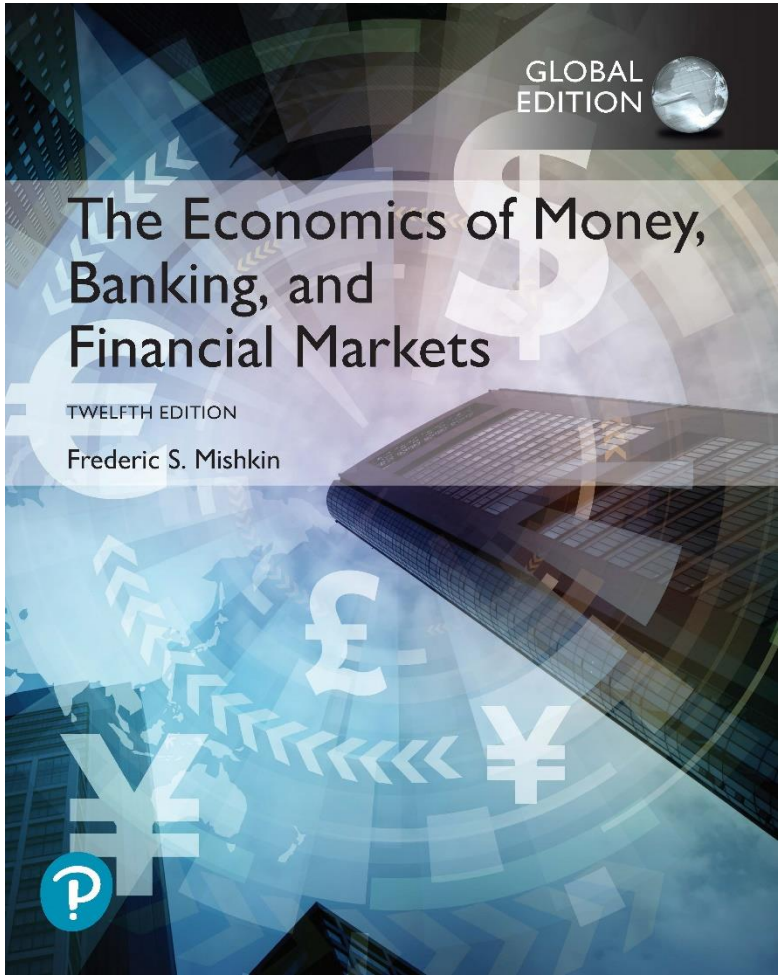
The Economics of Money, Banking, and Financial Markets

Twelfth Edition, Global Edition



Chapter 3

What Is Money?

Preview

- In this chapter, we develop precise definitions by exploring the functions of money, looking at why and how it promotes economic efficiency, tracing how its forms have evolved over time, and examining how money is currently measured.

Learning Objectives

- Describe what money is
- List and summarize the functions of money
- Identify different types of payment systems
- Compare and contrast the M1 and M2 money supplies

Meaning of Money (1 of 2)

- Money (or the “money supply”): anything that is generally accepted as payment for goods or services or in the repayment of debts.
- A broad definition

Meaning of Money (2 of 2)

- Money (a stock concept) is different from:
  - **Wealth**: the total collection of pieces of property that serve to store value
  - **Income**: flow of earnings per unit of time (a flow concept)

Functions of Money (1 of 2)

- **Medium of Exchange**:
  - Eliminates the trouble of finding a double coincidence of needs (reduces transaction costs)
  - Promotes specialization
- A medium of exchange must:
  - be easily standardized
  - be widely accepted
  - be divisible
  - be easy to carry
  - not deteriorate quickly

Functions of Money (2 of 2)

- **Unit of Account**:
  - Used to measure value in the economy
  - Reduces transaction costs
- **Store of Value**:
  - Used to save purchasing power over time.
  - Other assets also serve this function.
  - Money is the most liquid of all assets but loses value during inflation.

Evolution of the Payments System (1 of 2)

- **Commodity Money**: valuable, easily standardized, and divisible commodities (e.g. precious metals, cigarettes)
- **Fiat Money**: paper money decreed by governments as legal tender

Evolution of the Payments System (2 of 2)

- **Checks**: an instruction to your bank to transfer money from your account
- **Electronic Payment** (e.g. online bill pay)
- **E-Money** (electronic money):
  - **Debit card**
  - Stored-value card (**smart card**)
  - **E-cash**

Are We Headed for a Cashless Society?

- Predictions of a cashless society have been around for decades, but they have not come to fruition.
- Although e-money might be more convenient and efficient than a payments system based on paper, several factors work against the disappearance of the paper system.
- However, the use of e-money will likely still increase in the future.

Will Bitcoin Become the Money of the Future?

- Bitcoin is a type of electronic money created in 2009.
- By “mining,” Bitcoin is created by decentralized users when they use their computing power to verify and process transactions.
- Although Bitcoin functions as a medium of exchange, it is unlikely to become the money of the future because it performs less well as a unit of account and a store of value.

Measuring Money (1 of 2)

- How do we measure money? Which particular assets can be called “money”?
- Construct **monetary aggregates** using the concept of liquidity:
  - **M1** (most liquid assets) = currency + traveler’s checks + demand deposits + other checkable deposits

Measuring Money (2 of 2)

- **M2** (adds to M1 other assets that are not so liquid) = M1 + small denomination time deposits + savings deposits and money market deposit accounts + money market mutual fund shares

The Federal Reserve's Monetary Aggregates

(1 of 3)

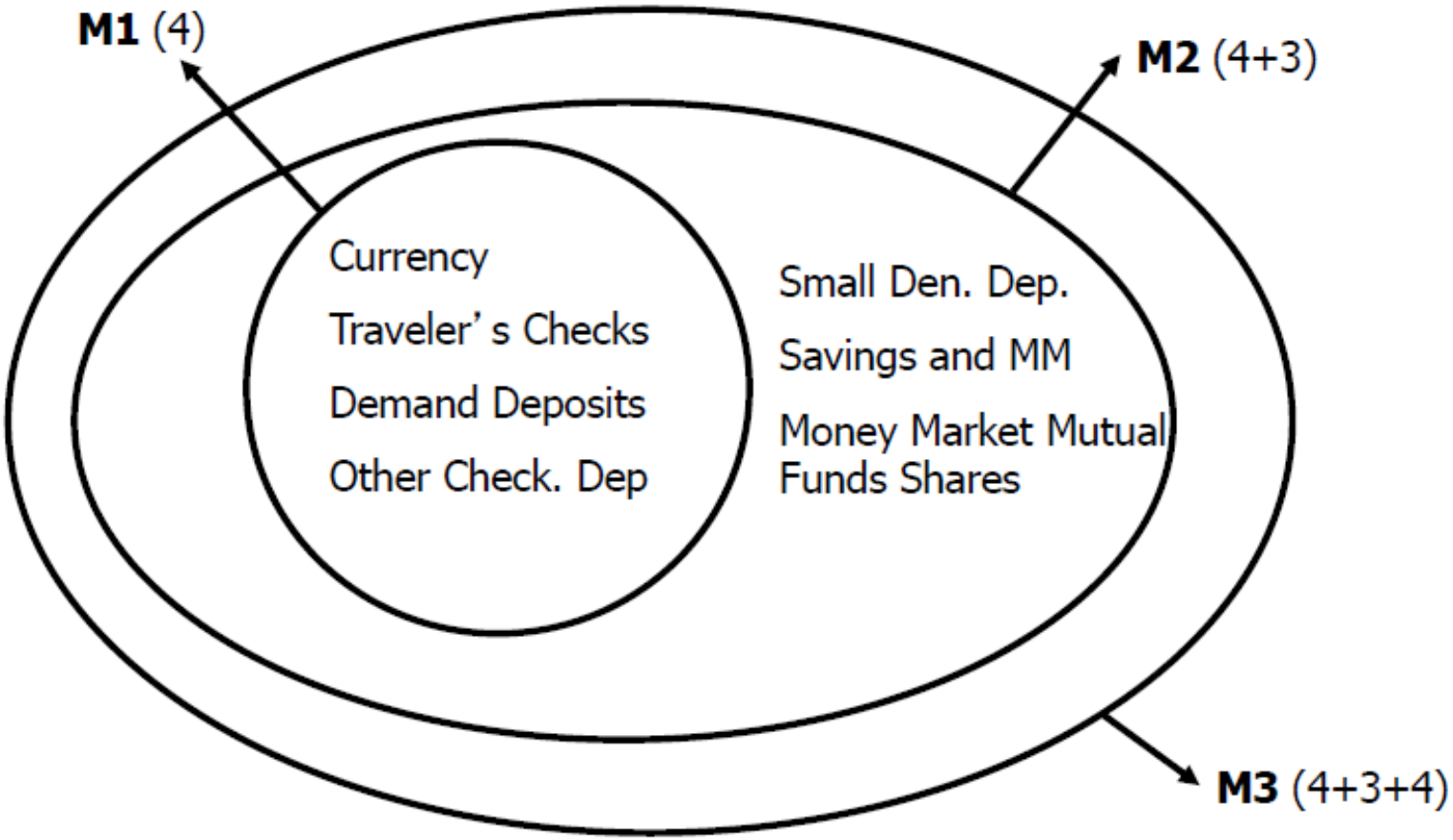
Table 1 Measures of the Monetary Aggregates

	Value as of July 3, 2017 ($ billions)
M1 = Currency	1,481.5
+ Traveler's checks	2.0
+ Demand deposits	1,501.5
+ Other checkable deposits	574.8
Total M1	3,559.8
M2 = M1	
+ Small-denomination time deposits	357.7
+ Savings deposits and money market deposit accounts	8,923.9
+ Money market mutual fund shares (retail)	673.7
Total M2	13,515.1

Source: Federal Reserve Statistical Release, H.6, Money Stock Measures: https://www.federalreserve.gov/releases/H6/current.

The Federal Reserve's Monetary Aggregates

(2 of 3)



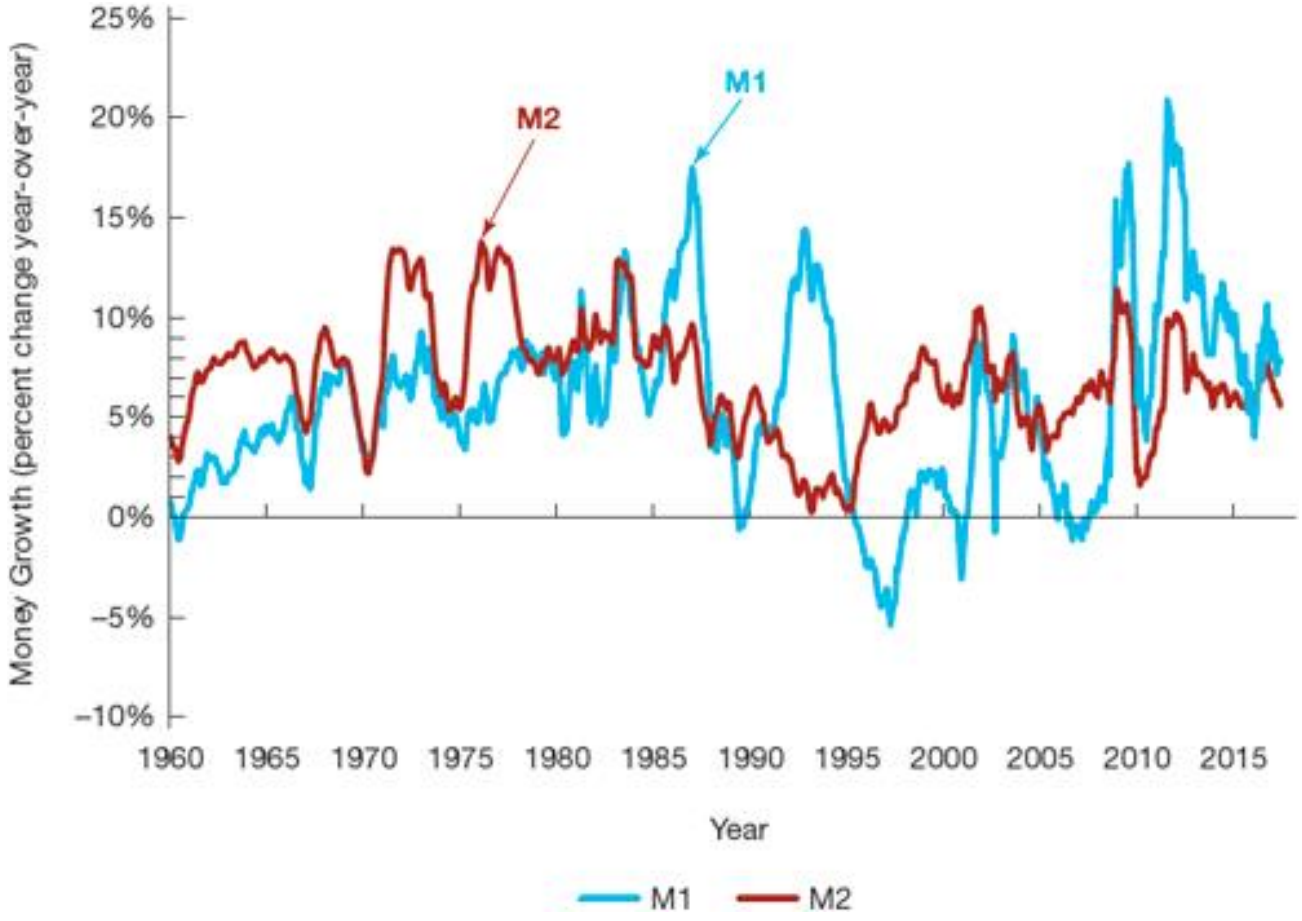
The Federal Reserve's Monetary Aggregates
(3 of 3)

- M1 versus M2: Does it matter which measure of money is considered?
- M1 and M2 can move in different directions in the short run (see figure).
- Conclusion: the choice of monetary aggregate is important for policymakers.

Where Are All the U.S. Dollars?

- The more than $4,500 of U.S. currency held per person in the United States is a surprisingly large number.
- Where are all these dollars and who is holding them?
  - Criminals
  - Foreigners

Figure 1 Growth Rates of the M1 and M2 Aggregates, 1960–2017



Source: Federal Reserve Bank of St. Louis, FRED database: http://research.stlouisfed.org/fred2